

DATA309 Project Description: Diabetes Prediction

Diabetes Prediction

Project Overview

The associated `diabetes_prediction_dataset.csv` file contains health and demographic data of 100,000 patients diagnosed with or without diabetes. The dataset consists of nine features: age, gender, body mass index (BMI), hypertension, heart disease, smoking history, Glycosylated Hemoglobin (HbA1c) level, blood glucose level, and diabetes status. Your task is to use this dataset to build and compare machine learning models, including Artificial Neural Networks (ANN) and traditional logistic regression, to predict (a) the level of glycosylated hemoglobin (HbA1c) and (b) the likelihood of diabetes in patients based on the variables presented. Additionally, you will investigate differences in predictions and feature impacts between males and females and relate your findings to existing research.

Using machine learning models, you will predict and test:

- HbA1c level from a set of predictor variables in the data (the final combination of variables is your choice).
- Likelihood of diabetes from a set of predictor variables in the data (the final combination of variables is your choice).

Implementation

Data Collection

Obtain raw data from the following GitHub repository:

https://raw.githubusercontent.com/arinbasu/data309/refs/heads/main/diabetes_prediction_data.csv

You can use R, Python, or Julia to download the data into your data science environment.

Pre-processing

Pre-process the data and accomplish the following tasks:

1. Identify each data type (e.g., numerical, categorical).
2. Plot each variable after selecting appropriate visualization types (e.g., histograms for numerical variables, bar plots for categorical variables).
3. Provide a statistical summary of each variable (e.g., mean, median, standard deviation for numerical; frequency counts for categorical).
4. Group the data by diabetes status (with and without diabetes) and summarize the eight other variables based on this classification. Include a breakdown by gender to explore differences between males and females.
5. Generate appropriate plots, including comparative visualizations (e.g., boxplots, histograms) to highlight differences in variables like age, BMI, HbA1c, and blood glucose level between males and females within each diabetes group.

Analysis Workflow

1. **ANN for HbA1c Prediction:** Develop an Artificial Neural Network using an appropriate ANN tool (e.g., “Pytorch” or “keras” for Python, “neuralnet” in R, or “flux” in Julia) to predict HbA1c values from a selected set of predictor variables (at minimum, include “Age”, “BMI”, and “Blood Glucose Level”) using a linear regression approach. Split the dataset (80:20) into training and testing data, set up the neural network, and provide evidence of model fit. Select the best optimizer and activation function for the model and justify your choices. Evaluate potential over-fitting by monitoring training and validation loss curves and applying techniques like regularization or early stopping.
2. **ANN for Diabetes Likelihood:** Develop an Artificial Neural Network using an appropriate ANN tool to predict the likelihood of diabetes from a selected set of predictor variables (at minimum, include “Age”, “BMI”, and “Blood Glucose Level”) using a logistic regression approach. Split the dataset (80:20) into training and testing data, set up the neural network, and provide evidence of model fit (e.g., accuracy, loss). Select the best optimizer and activation function for the model and justify your choices. Evaluate potential over-fitting using techniques like dropout.
3. **Traditional Logistic Regression Model:** Using scikit-learn in Python (or equivalent libraries in R or Julia, such as “glm” in R or “GLM” in Julia), build a traditional logistic regression model to predict the likelihood of diabetes using the same predictor variables selected for the ANN in task 2. Split the dataset (80:20) into training and testing data, fit the model, and evaluate its performance using metrics such as accuracy, sensitivity, specificity, and Area Under the ROC Curve (AUC-ROC).
4. **Model Comparison:** Compare the performance of the ANN and traditional logistic regression models for predicting diabetes likelihood. Report differences in performance

metrics (e.g., accuracy, sensitivity, specificity, AUC-ROC) and discuss potential reasons for discrepancies, such as model complexity, feature interactions, or data characteristics.

5. **Gender-Based Analysis:** For both the ANN and traditional logistic regression models predicting diabetes likelihood, analyze differences in model performance and feature importance (e.g., coefficients in logistic regression, feature attribution in ANN) between males and females. Split the dataset by gender, train and test the models separately for each group, and report any differences in predictive performance or key predictors. Visualize these differences using appropriate plots (e.g., coefficient plots, feature importance plots).
6. **Relate to Existing Research:** In your report, discuss how your findings align with or differ from existing research on diabetes prediction. Cite at least three relevant studies (e.g., from PubMed, Google Scholar, or other academic sources) that use similar predictors or models to predict HbA1c or diabetes likelihood. Highlight how your choice of variables, model performance, or gender-based findings compare to these studies and discuss potential implications for clinical practice or future research.

Evaluation

- For the HbA1c prediction model (ANN), evaluate and report performance metrics (e.g., Mean Squared Error, R-squared) and show how the predicted data compare to the test dataset. Discuss evidence of model fit and strategies used to prevent overtraining (e.g., regularization, early stopping, validation loss monitoring).
- For the diabetes likelihood models (ANN and traditional logistic regression), prepare confusion matrices and report sensitivity, specificity, accuracy, and AUC-ROC. Compare the performance of the ANN and logistic regression models, highlighting strengths and weaknesses of each approach.
- For the gender-based analysis, report differences in model performance (e.g., sensitivity, specificity, AUC-ROC) and key predictors between males and females. Discuss potential reasons for these differences, such as biological or demographic factors.
- In your report, synthesize your findings with existing research, discussing how your results contribute to the understanding of diabetes prediction and any limitations of your approach.

Tools and Technology

You can use Julia, Python, or R for this project. For Python, scikit-learn is recommended for the traditional logistic regression model, and Pytorch or keras (or similar) for ANN implementation. For R, use “glm” for logistic regression and “neuralnet” for ANN. For Julia, use “GLM” for logistic regression and “flux” for ANN.

You may use Google Colab for reading data and conducting analyses.

Resources

- **Dataset:** Download from https://raw.githubusercontent.com/arinbasu/data309/refs/heads/main/diabetes_prediction_data.csv
- **RStudio:** <https://rstudio.cloud>
- **JuliaHub:** <https://www.juliahub.com>
- **Google Colab Notebooks:** <https://colab.research.google.com>
- **Research References:** Use academic databases like PubMed or Google Scholar to find relevant studies on diabetes prediction.